

Part B-1

1. Excellence

1.1 *Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)*

Problem and overarching aim. European agriculture faces increasing water constraints: approximately 30% of EU territory experiences seasonal water scarcity in an average year, while climate change is expected to intensify drought frequency, severity and seasonal freshwater fluctuations¹. FIELDWISE asks: **can a physiologically grounded plant water-stress signal remain valid across seasons, environments and sensing scales, survive prospective validation under real commercial-field conditions, and be translated into crop-configurable irrigation decision support?** Processing tomato is the **development and validation crop**, supported by an unusually rich five-season (2022–2026) archive of physiological, soil, meteorological and proximal-spectral observations. An end-of-project non-academic placement completes inter-sectoral translation.

Pertinence. Three linked scientific gaps motivate the objectives: (i) insufficient physiological grounding of water-stress indicators; (ii) limited evidence of predictive validity across unseen years, environments and sensing scales; and (iii) weak translation into operational irrigation decision support. Spectral and environmental variables frequently serve as stress proxies without demonstrated consistency with plant physiological responses². Existing tomato approaches already supply thermal/CWSI stress assessment³, soil-water, evapotranspiration and AquaCrop-based scheduling⁴ and machine-learning irrigation support for processing tomato⁵; none has been tested as a model frozen before deployment and carried, with its uncertainty, across years, an independent field and the proximal-to-satellite scale, which is the evidence FIELDWISE adds.

The four objectives, and why they are measurable and verifiable. O1 — Harmonise the 2022–2026 MATE archive and establish the water-stress reference and validation protocol. The five-season archive of physiological, soil-water, meteorological and hyperspectral observations under contrasting, AquaCrop-derived irrigation treatments^{6,7} is harmonised, and canopy spectra are converted to synthetic Sentinel-2 predictors only where wavelength coverage and calibration records support it (Section 1.2). Outputs: harmonised database and predictor-availability map (D1.1, M2); stress-reference, feature-selection and physiological-validation protocol (D1.2, M3). **O2 — Develop an interpretable and uncertainty-aware probabilistic water-stress model.** A regularised logistic regression on a pre-specified predictor set, benchmarked against constrained gradient-boosted trees, predicts water-stress probability and operational stress status from Sentinel-compatible indices, soil-water and meteorological variables against an independently measured physiological reference. Output: model, benchmark, predictor definitions and model card frozen at M3 (D1.3). **O3 — Prospectively validate the frozen model under unseen commercial-field conditions over two processing-tomato seasons.** A replicated design (three irrigation regimes in four blocks, Section 1.2) in a commercial field with valid Sentinel-2 treatment areas, with coordinated physiology verifying the plant response independently. Outputs: prospective validation dataset (D2.1, M20); transferability and uncertainty report (D2.2, M21). **O4 — Translate validated outputs into the DrR web MVP and assess their operational relevance.** Building on the fellow's pre-existing desktop DrR prototype, with Krumatic engineering and AgroVIR requirements, feedback and final non-academic evaluation, only validated outputs enter the prototype. Outputs: web MVP and readiness dossier (D3.1–D3.2, M24); operational/interoperability assessment and roadmap (D4.1–D4.2, M30).

Measurability and verifiability. Model development within the historical archive will use blocked, leakage-safe unseen-year validation. The stress-reference definition, predictors, feature-selection rules, evaluation metrics and permitted recalibration strategy will be pre-specified in D1.2. The frozen model will subsequently be tested

1 EEA (2025), Water scarcity conditions in Europe, indicator, 28 Nov. 2025, www.eea.europa.eu/en/analysis/indicators/use-of-freshwater-resources-in-europe-1

2 Gerhards M. et al. (2019), Challenges and Future Perspectives of Multi-/Hyperspectral Thermal Infrared Remote Sensing for Crop Water-Stress Detection: A Review, *Remote Sensing* 11, 1240.

3 Alordzinu, K.E., Li, J., Lan, Y. et al. (2021), Rapid Estimation of Crop Water Stress Index on Tomato Growth, *Sensors* 21, 5142. doi:10.3390/s21155142

4 Allen, R.G. et al. (1998), *Crop Evapotranspiration*, FAO Irrigation and Drainage Paper 56, FAO, Rome.

5 Martelli, A. et al. (2025), Smart irrigation for management of processing tomato: a machine learning approach, *Irrigation Science* 43, 1407–1424. doi:10.1007/s00271-024-00993-9

6 Takács S. et al. (2020), Influence of Water Stress Levels on the Yield and Lycopene Content of Tomato, *Water* 12, 2165.

7 Takács S. et al. (2021), Performance Evaluation of AquaCrop Model in Processing Tomato Biomass, Fruit Yield and Water Stress Indicator Modelling, *Water* 13, 3587.

prospectively over two unseen commercial-field seasons. Performance will be evaluated using balanced accuracy, F1 and ROC-AUC for stress detection; reliability and Brier score for probabilistic calibration; and lead time and false-alarm rate for operational value. Regression metrics will be used only if the archive supports a defensible continuous stress-severity reference. Freezing the model and predictor definitions at M3 prevents retrospective optimisation against prospective outcomes and makes the central hypothesis directly falsifiable: whether water-stress relationships learned from controlled experimental observations remain valid when corresponding predictors are obtained from actual Sentinel-2 observations under commercial-field conditions.

Realistically achievable. The existing five-season MATE processing-tomato archive and controlled irrigation experiments support feasibility, and the fellow's 2025 MATE training stays and co-authored processing-tomato manuscript (Section 1.4) mean the M1 archive audit starts from familiarity rather than first contact. Prospective validation covers one commercial environment, two growing seasons and controlled irrigation contrasts; DrR builds on an existing prototype. In-action validation remains limited to processing tomato; a crop-transfer specification and subsequent post-project replication and knowledge transfer through MVCRI address additional crops.

Beyond the state of the art, and the extent of the ambition. FIELDWISE challenges four assumptions: vegetation indices themselves represent water stress; proximal and satellite observations are interchangeable; internal validation demonstrates field transferability; and one crop's model transfers unchanged to another. Its ambition is a physiologically validated, uncertainty-aware transfer framework, not a new machine-learning algorithm, testing whether relationships learned from controlled experiments remain valid across unseen years, sensing scales and commercial fields. Three citable advances are: (1) a probabilistic processing-tomato water-stress model documenting calibration, uncertainty and blocked unseen-year performance; (2) a prospective experimental-to-commercial-field transfer gap quantified with a model frozen before deployment; and (3) a specification separating reusable modelling components from sensing-scale and crop-specific elements requiring recalibration. The new knowledge quantifies **whether, how far and under what conditions predictive validity survives transfer from controlled experiments to operational satellite-based monitoring.**

1.2 *Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices).*

Overall methodology: concepts, models and assumptions. Four linked criteria evaluate water-stress prediction: physiological validity, predictive performance, transferability and management relevance. Irrigation treatments and AquaCrop water balances supply the imposed-deficit context and agronomic comparators, not the outcome, which would be circular. D1.2 fixes the stress reference before any fitting. Its primary axis is stomatal conductance relative to stage- and campaign-matched well-watered controls, corroborated by leaf water status or another consistently available independent marker deviating beyond its repeatability limit; chlorophyll fluorescence is complementary evidence, and a normal Fv/Fm alone does not label a plant unstressed. Reference limits derive from historical controls and measurement repeatability, never from model accuracy or prospective outcomes; ambiguous or confounded area-dates are classified indeterminate and retained in sensitivity reporting. Without a common two-marker reference across the archive, one continuous physiological endpoint becomes the primary reference and other measurements corroborate it.

Stage 1 (WP1) harmonises the 2022–2026 MATE archive into a QC-controlled common-variable matrix with explicit missingness and availability maps. An M1 audit records per season the instrument, wavelength coverage, calibration, footprint, plot geometry, canopy, treatments, dates and matched physiology (D1.1). Historical plots are too small for plot-level Sentinel-2 retrieval, so calibrated canopy reflectance is integrated with versioned Sentinel-2 spectral-response functions into synthetic band reflectance, indices are computed after band integration, and bands outside the instrument's range are dropped, not reconstructed. Proximal white/dark-reference correction stays separate from satellite atmospheric processing; Sentinel-2 inputs are cloud/shadow-masked Level-2A surface reflectance on a common 20 m grid where 20 m bands are used, with interior-pixel rules ($\geq 90\%$ treatment cover per pixel, ≥ 4 valid pixels per area and date). A predictor is retained only if it passes the D1.2 engineering screen (median absolute index discrepancy ≤ 0.25 historical standard deviations, $\leq 10\%$ change in stress classifications on a locked sensitivity set); otherwise the model uses the defensible common predictors alone, a pre-specified soil-water/meteorological baseline is retained and the failed spectral transfer is reported. The primary model is a regularised logistic regression on a limited pre-specified predictor set, benchmarked against constrained gradient-boosted trees; outer leave-one-year-out validation across the five seasons tests temporal transfer, with imputation, scaling, feature selection, tuning and calibration fitted only inside training folds under year- and group-aware inner

splits, uncertainty treated group-aware or hierarchically (each treatment area's observations kept together; seasons reported separately and pooled), and abstention or a limited-evidence flag for out-of-domain or low-quality observations. The pipeline is selected under a declared historical-data rule, refitted on historical data only and frozen with its model card at M3 (D1.3), before use of any prospective-field outcome. D1.2 also fixes the decision criteria: balanced accuracy ≥ 0.70 with a demonstrated advantage over a locked agronomic comparator, positive Brier skill against a frozen reference model, an alert threshold set on historical predictions for sensitivity ≥ 0.80 at a false-positive rate ≤ 0.20 , and interval-bounded lead time where observation frequency does not support a point estimate; failing a criterion restricts the model to research-only use and the MVP accordingly, and unchanged-model results are always reported before any separately versioned recalibration.

Stage 2 (WP2) prospectively tests environmental and sensing-scale transfer over two processing-tomato seasons in an independent commercial field. The adopted design, implemented subject to the field-readiness gate, has three irrigation regimes (well-watered, deficit-irrigated, unirrigated) in four spatial blocks: 12 separately managed, randomised treatment areas per season of about 80×80 m, a 20 m inward buffer leaving a 40×40 m interior core, and about 7.68 ha before access and isolation land. Twelve physiological campaigns per season across canopy development, flowering/fruit set, fruit filling and ripening, with temperature/VPD, genotype, time of day and competing stresses logged, sample five spatially distributed plants per area. The separately managed area is the experimental unit (plants, pixels and repeated dates are not independent replicates): up to 288 area-date records before missingness, about 202 area-date pairs under 30% attrition, and two years on one field give temporal replication, not two independent environments. The frozen model uses actual Sentinel-2 indices, soil-water and meteorological observations; coordinated physiology independently verifies realised plant stress. Primary performance precedes any recalibration; subsequent limited recalibration is secondary and pre-specified. The transfer gap quantifies predictive validity retained from controlled experiments to genuine satellite-scale field monitoring (**D2.1–D2.2, M20–M21**). Crop transfer is addressed separately, distinguishing reusable components from those requiring crop-specific recalibration and revalidation. Paired proximal canopy spectra on selected campaigns, aligned with Sentinel-2 observations and physiology, diagnose sensing discrepancies without tuning the frozen model; if sensing and environmental effects cannot be separated, the combined gap is reported and the claim narrowed.

Stage 3 (WP3–WP4) operationalises only validated functions: the fellow's pre-existing DrR – Digital Agronomist research prototype is extended with validated scientific functionality, Krumatic implements it and AgroVIR evaluates operational usability and interoperability.

Methodological challenges and mitigation. (i) A year-by-variable availability matrix and only defensible common predictors manage inter-annual archive heterogeneity; (ii) pre-model spectral harmonisation and prospective scale-transfer testing link hyperspectral and satellite observations without assuming interchangeability; (iii) blocked unseen-year validation avoids optimistic random-split generalisation; (iv) a model frozen before deployment tests commercial-field transfer, reporting performance before recalibration; (v) sufficiently large treatment areas and valid interior observations reduce Sentinel mixed-pixel risk; and (vi) parsimony, uncertainty assessment, lead time and false-alarm rate control over-fitting. A predefined site-contingency plan protects two-season field validation.

Integration of methods and disciplines. HUN-REN ATK supplies plant-stress physiology and the scientific gates (Prof. Janda) and statistical design, calibration and leakage checks (Dr Hollós); ELTE sensor harmonisation and sensing-scale transfer (Prof. Jung); MATE the controlled-irrigation archive and agronomic review of the irrigation protocol before lock (Dr Takács); Krumatic/AgroVIR software-engineering and operational farm-management perspectives. The fellow integrates these disciplines: the biological reference constrains model interpretation, the protocol constrains development, and the validated predictor specification governs operational field inputs.

Gender dimension and other diversity aspects. A sex/gender dimension is not relevant to the biological research content, which concerns processing-tomato water-stress responses and involves no human or animal biological data. Diversity becomes relevant only in end-user interaction: usability feedback will consider differences in farm context and digital literacy. Any grower feedback will follow informed-consent, withdrawal, data-minimisation and coded/anonymised-processing principles.

Open science practices. FIELDWISE follows the principle “as open as possible, as closed as necessary.” The stress-reference rules, predictor definitions, metrics, acceptance criteria and freeze record (D1.2) are registered with a timestamp before model fitting and deposited publicly with their metadata; D1.3 carries a versioned model card (intended use, validation, uncertainty, limitations). Deposits go to Zenodo or an equivalent repository with persistent identifiers; separable research code is released under BSD-3-Clause where ownership and dependencies permit, and metadata, access terms and synthetic examples are published where raw or background data cannot be.

The prospective dataset and transfer analysis (D2.1–D2.2) are released in rights-cleared, non-identifying form; manuscripts are posted as preprints on submission where permitted, and successful and unsuccessful transfer results are both reported. A FAIR Data Management Plan is completed within six months. The scientific evidence package stays separate from the proprietary DrR, contractor code and FMIS layers; a 30-day project disclosure review, subject to mandatory institutional clearance, bounds the assessment of protectable results before release.

1.3 *Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host*

Supervisory architecture and qualifications. Primary supervision at the recruiting organisation, HUN-REN ATK, is provided by **Prof. Tibor Janda**, Head of the Plant Physiology and Metabolomics Department at the Agricultural Institute, accountable for the physiological reference and the scientific gates. His plant-physiology, drought/abiotic-stress and stress-metabolomics expertise anchors the physiological-validation framework and the measurement programme. His supervision record is documented: doctoral-school core member since 2010 and council member since 2013, lecturer in plant physiological investigation methods (1997–2012), co-supervisor of a 2023 doctoral dissertation at Széchenyi István University, and research stays at CEA Saclay and Ben-Gurion University. **Dr Roland Hollós** (ATK) is accountable for statistical design, calibration and leakage checks in the predictive-modelling and machine-learning strand⁸. **Prof. András Jung (ELTE, Institute of Cartography and Geoinformatics)** provides specialist co-supervision in field spectroscopy, hyperspectral imaging, geoinformatics, satellite Earth observation and sensor harmonisation⁹, is a doctoral-programme head at ELTE and has research and teaching experience at Halle-Wittenberg, Leipzig and Ulm. MATE contributes the five-season controlled-irrigation processing-tomato archive and agronomic expertise through **Dr Sándor Takács**, including AquaCrop-based irrigation and crop-water-stress methodology.

Structured supervision and governance. Supervision comprises regular one-to-one meetings with Prof. Janda, dedicated modelling/EO technical sessions with specialists, monthly written progress monitoring and quarterly Career Development Plan reviews. The **Project Steering Group**—fellow, primary supervisor, specialist co-supervisor/modelling expert and relevant partner contacts—reviews protocol/model freeze, prospective-field implementation, transfer-validation completion, MVP readiness and the non-academic placement. Pre-defined decision responsibilities and a two-step conflict-resolution route ensure continuity while protecting the fellow's growing scientific independence.

Planned training activities and ELTE secondment. Training is embedded in implementation and assessed by output: M1–M3 at ELTE (Prof. Jung), a supervised module on the ELTE subjects Hyperspectral Imaging and Field Spectroscopy and Remote Sensing Methods and Technologies (output: reproducible band-integration/QC notebook); M1–M3 at ATK (Prof. Janda, Dr Takács), physiological-reference and irrigation-design practical (reviewed reference protocol, repeatability exercise); M1–M3 and M10–M12 at ATK (Dr Hollós), blocked validation, calibration and uncertainty practical (rerun historical folds, model-card review); M4–M9, host research-support staff, reproducible research, FAIR data, responsible generative-AI use and IP (release package, DMP); M16–M21 at ELTE, paired-sensing and transfer-diagnostics clinic; M22–M30, ATK support and AgroVIR, grant-writing clinic and FMIS evaluation (Section 2.1). The 12-month ELTE secondment (M1–M3, M10–M12, M16–M21) hosts the EO strands. During M1–M3, primary supervision continues through weekly Janda–fellow meetings, fortnightly joint physiology/EO/modelling sessions, shared versioned records and a joint sign-off before protocol lock and model freeze; a documented continuity arrangement designates a deputy for prolonged absence. Teaching: an ATK/ELTE practical on physiological validation by M12 and a transfer-analysis practical by M24; guest lectures at MATE or the Agricultural University–Plovdiv when scheduled, a host seminar otherwise. The **Career Development Plan (D5.1, M3)** consolidates the programme and the transferable-skills portfolio (Section 2.1), reviewed quarterly (M6–M30) with transition reviews at M12/M21/M24/M27.

Two-way transfer of knowledge. HUN-REN ATK contributes advanced abiotic-stress physiology, phenotyping and modelling; ELTE established hyperspectral-sensing, geoinformatics and satellite-EO expertise. The fellow contributes drought-stress physiology, remotely monitored plant-stress responses, field physiological phenotyping and interpretation under contrasting water availability, strengthening stress biology's translation into irrigation decision support. Joint supervision, the distributed ELTE secondment, shared MATE-archive analysis, prospective field validation, teaching and co-authored publications reinforce exchange.

⁸ Hollós R. et al. (2026), Meta-modelling of carbon fluxes from crop and grassland multi-model outputs, Geoscientific Model Development 19, 4385–4438.

⁹ Jung A. et al. (2010), White-reference based post-correction method for multi-source spectral libraries, PFG 2010(5), 363–369.

Rationale and added value of the non-academic placement. The six-month AgroVIR placement (M25–M30) is an integral inter-sectoral component of FIELDWISE, providing competences not fully available academically: DrR web-MVP operational evaluation, FMIS requirements engineering, interoperability assessment, user-oriented testing and exploitation planning. AgroVIR supplies requirements feedback during the fellowship; the placement adds intensive exposure to production agricultural software and decision-support workflows. Structured technical reviews assess usability, data-flow requirements, missing-data behaviour, interoperability and the pathway toward future FMIS integration. Confidentiality, DrR background access and result ownership are defined in advance under the Grant Agreement (Section 2.2).

1.4 *Quality and appropriateness of the researcher's professional experience, competences and skills*

A profile matched to the project's interdisciplinary chain. Dr. Rositsa Cholakova (publishing before 2020 as Rositsa Cholakova-Bimbalova) holds BSc and MSc degrees in Plant Protection and a PhD in Plant Physiology (2020, Agricultural University of Plovdiv). She was Assistant Professor in Plant Physiology and Plant Stress Physiology in Plovdiv (2015–2020) and is currently Chief Assistant Professor at the Maritsa Vegetable Crops Research Institute. **Physiology:** her published maize stress-physiology work and documented methods (chlorophyll fluorescence, stomatal conductance, pigment and leaf water status, field stress assessment) equip her to define and measure the independent physiological reference. **Earth observation and data science:** a COST PANGEOS Short-Term Scientific Mission at MATE (10 days, 2025), Erasmus+ RGB/hyperspectral training at MATE (14 days, 2025) and a three-month ELTE traineeship in geoinformatics and remote sensing (2026), with working use of Python, R and EO tools, prepare her for the archive audit, band integration and Sentinel-2 processing. **Recent crop work:** a 2025 proceedings paper on remote-sensing monitoring of watermelon under different water regimes and two 2026 manuscripts, under review, on UAV-based evapotranspiration in tomato fields and on proximal hyperspectral signatures of water stress in processing tomato with physiological and soil-water validation, the latter co-authored with Dr Takács (Part B-2, Section 4). She holds an EU A1/A3 drone-pilot qualification and developed **DrR – Digital Agronomist**, a pre-existing desktop research prototype that FIELDWISE extends only with validated functions. Five years of university teaching in Bulgarian and English underpin the teaching strand. **Fit and development potential.** The fellowship addresses the next step in her progression: advanced predictive modelling, uncertainty analysis, leakage-safe validation and transfer across years, sensing scales and commercial-field conditions, deepened through the distributed ELTE secondment and specialist supervision at HUN-REN ATK. Following a full-time-parenting career break (2021–2024), she has re-entered research with a focused transition toward an independent **Agricultural Data Scientist working at the interface of plant physiology, Earth observation and irrigation decision support.**

2. Impact

2.1 *Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his/her skills development*

FIELDWISE is designed to consolidate the fellow's transition from plant-stress physiologist and agronomist to an **independent Agricultural Data Scientist working at the interface of plant physiology, Earth observation, predictive modelling and irrigation decision support.**

Two realistic post-fellowship trajectories anchor this development. The first is an independent research/data-science role in European academia, research organisations or agri-technology. The second is development of an independent research and innovation line in Bulgaria, extending the FIELDWISE methodology to further irrigated high-value crops and transferring the acquired competences through MVCRI. Progress is tracked through the Career Development Plan and career KPI **K11**.

Measure	Skills and employability mechanism	Evidence
Career Development Plan	Co-developed with the supervisors; translates career objectives into technical training, publications, teaching, networking, leadership and open-science actions, with progress reviewed at major project transitions.	D5.1 (M3); quarterly reviews M6–M30; transition reviews M12/M21/M24/M27; K11
ATK modelling training and 12-month ELTE secondment	ATK: plant-stress interpretation, experimental design, interpretable modelling, leakage-safe validation, calibration and uncertainty. ELTE (M1–M3, M10–M12, M16–M21): hyperspectral-to-Sentinel harmonisation, geoinformatics, EO processing and sensing-scale transfer, applied first to historical predictors and then to real Sentinel-2 field observations.	Supervision records; model card; D1.3; secondment outputs; CDP

Measure	Skills and employability mechanism	Evidence
Transferable-skills portfolio	Research integrity, FAIR/open science, responsible AI, grant writing, IP, project management and science communication, exercised through FIELDWISE. Independence measures: lead a methods clinic by M12; prepare a grant concept and mock-reviewed application by M21–M24; mentor a junior researcher or student on a reproducible analysis task in M10–M21 (peer-learning group if no mentee).	CDP-tracked: clinic (M12), reviewed application (M24), mentoring record (M21)
Non-academic placement	Six months at AgroVIR (M25–M30) developing competence in FMIS requirements, interoperability, user-oriented evaluation and exploitation planning.	D4.1–D4.2 (M30)
Scientific visibility and leadership	Co-authored publications, conference presentations, seminars and guest lectures at MATE and the Agricultural University–Plovdiv, and the COST Action CA22136 PANGEOS contact base, build visibility, networks and future supervision capacity.	≥2 manuscripts; conference/teaching outputs

By completion, the fellow will have taken one research concept through biological definition, historical-data harmonisation, predictive modelling, unseen-year validation, prospective satellite-based field validation, software translation and industrial evaluation, a combination that strengthens employability across academic plant/EO research, agricultural data science and agri-technology.

2.2 *Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities*

FIELDWISE uses a targeted strategy proportionate to a single-fellow action, progressing from scientific dissemination → practitioner communication → operational and entrepreneurial exploitation.

Strand	Target groups	Measures and outputs	Evidence/timing
Dissemination	Plant physiology, crop modelling, RS/EO, precision agriculture; students	≥2 peer-reviewed manuscripts, possible third; model card; hyperspectral-to-Sentinel/transfer methodology; prospective transfer performance published, successful or not; EGU/ECPA and an Irrigation Science paper or ISHS irrigation symposium; registered protocol, metadata, separable code and non-sensitive outputs released with persistent identifiers.	D1.3 M3; D2.1 M20; D2.2 M21
Communication	Growers and advisors (HU/BG); wider public; HU/BG students and practitioners	Growers/advisors: what an uncertainty-qualified alert means and when no advice is justified; practitioner brief (M12), two demonstrations at the AgroVIR evaluation (M25–M29), one workshop of about 20 participants (M29–M30). Public: why plants, satellites and uncertainty are checked together before advice; three illustrated explainers (M3, M12, M24). Students/practitioners (HU/BG): which methods transfer and which need local validation; one reusable tutorial and Hungarian/Bulgarian summaries (M21–M24). Claims restricted to measured evidence.	Attendance, feedback, views and downloads; reported in D5.3
Exploitation	FMIS/agri-software sector, growers, technology-transfer actors	Maturation of the fellow's pre-existing DrR desktop prototype into a validated web MVP . FIELDWISE provides validated scientific/EO functionality; Krumatic provides remunerated software engineering; AgroVIR provides requirements and operational evaluation. Post-MSCA routes include FMIS integration, MVCRI crop extension and HUN-REN/HUNRENTech-supported licensing or researcher-led start-up/spin-off exploration.	D3.1–D3.2 M24; D4.1–D4.2 M30
IP and knowledge management	Fellow, beneficiary and contributing organisations	Separate rights layers for DrR background / FIELDWISE-generated results / institutional datasets / contracted code . A dated DrR declaration establishes the pre-action baseline. Ownership/licensing of the new web software and source code is defined through the relevant HUN-REN ATK–Krumatic agreement; AgroVIR receives only access required for evaluation.	Pre-action agreements; maintained IP inventory

Exploitation pathway. DrR enters FIELDWISE as the fellow's pre-existing, independently developed research prototype. Only scientifically validated functions are transferred into the web MVP. At M24 the result is a validated MVP, followed by independent AgroVIR evaluation in M25–M30 before any claim of FMIS readiness or wider commercial deployment. FIELDWISE therefore delivers four exploitable result families: the validated stress-prediction workflow, the hyperspectral-to-Sentinel transfer methodology, the matured DrR web MVP, and the AgroVIR-informed FMIS-integration specification.

Reach and uptake are tracked through project-controlled evidence: ≥ 2 conference presentations and submitted manuscripts; structured evaluation with 6–10 growers or advisors and 2–3 FMIS or technical users (task completion, uncertainty interpretation, usability); two demonstrations and one workshop with recorded attendance and feedback; three explainers, one tutorial and two language summaries; the AgroVIR assessment; repository downloads. AgroVIR recruitment is an agreed target, not a guaranteed channel; attendance is not adoption.

2.3. The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

Impact is bounded to what the action demonstrates directly: whether physiologically grounded water-stress relationships remain valid across unseen years, sensing scales and two prospective seasons, and whether validated results support operational decisions.

Dimension	Contribution	Magnitude and importance
Scientific	Leakage-safe modelling links physiological evidence with spectral, soil-water and meteorological predictors; a frozen model is tested across unseen years and an independent commercial field.	Outputs: a documented model, quantified transfer performance and a specification separating reusable from crop-/scale-specific components, showing whether, how far and under which conditions predictive validity survives transfer .
Agricultural & environmental	The workflow provides stress probability, uncertainty and irrigation-attention information independently checked against plant physiology across two seasons.	The direct value is more defensible identification of when and where water stress requires attention . Warning lead time, false alarms, uncertainty and water–yield–quality relationships are measured prospectively; no numerical water-saving benefit is claimed in advance. Irrigation, yield and quality are recorded by treatment, not used to estimate DrR-caused savings.
Technological & economic	The fellow's pre-existing DrR desktop prototype is matured into a validated web MVP; Krumatic implements the software layer and AgroVIR evaluates usability and interoperability.	The pathway is auditable: pre-existing prototype → validated web MVP → independent operational assessment → post-MSCA FMIS/exploitation roadmap . This reduces the scientific and operational uncertainty required for later integration, licensing or researcher-led innovation.
European & societal	A Bulgarian researcher trained in Hungary across an academic host, a university secondment host and a non-academic FMIS company, with a potential MVCRI continuation route.	Serves the EU Water Resilience Strategy's efficiency and digitalisation priorities (its 10% economy-wide ambition is not a project target); delivers the call's fellowship outcomes: skills and employability, international/interdisciplinary/inter-sectoral experience, knowledge exchange, research-to-teaching transfer, each with an evidencing output.

Impact is measured through project-controlled evidence: discrimination and calibration, uncertainty, false alarms, warning lead time and prospective transfer performance, all pre-specified in D1.2 and reported before any limited recalibration. Three scales are kept distinct: direct delivery (one crop, one commercial environment over two seasons, one frozen model, one web MVP, one industrial evaluation); the translation channel, where AgroVIR reported in August 2025 that its software serves over 700,000 ha across Hungary and six other countries, an established distribution context that is neither FIELDWISE coverage nor irrigated-tomato area, adoption or water saving; and eligible uptake, the subset of that network with suitable crop, irrigation, field-size and data conditions, quantified during the placement (denominator, date, definition), with later adoption modelled only as an explicit scenario after validation and a separate adoption decision.

3. Quality and Efficiency of the Implementation

3.1 *Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages*

The fellow leads the scientific work at the sole beneficiary, HUN-REN ATK (ATK), over 24 months, followed by the AgroVIR placement (M25–M30). ELTE secondment blocks are M1–M3, M10–M12 and M16–M21 (12 months); they are locations of work within the allocated effort. MATE supplies archive/irrigation expertise; the producer hosts field trials; Krumatic supplies remunerated engineering.

WP1 — Data harmonisation, stress definition & model development | M1–M21 | 5.0 PM.

Lead: fellow/ATK; MATE and ELTE support. **T1.1** (M1–M2): audit and harmonise all five archive seasons. **T1.2** (M1–M3): harmonise predictors and lock the physiological reference, folds, metrics, sampling and recalibration rules before fitting. **T1.3** (M2–M3): build the pipeline; after protocol lock, fit and freeze the model/benchmark. **T1.4** (M10–M12; M16–M21): synthesise WP2 transfer diagnostics without changing the primary model.

D1.1 archive/QC map (M2); D1.2 protocol and D1.3 model/card (M3); D1.4 transfer specification (M21). K1–K4: five seasons audited; one locked protocol; reproducible, leakage-safe held-out-year results; one reusable/crop-specific transfer specification. Gates: MS1/MS3.

WP2 — Prospective field validation | M1–M21 | 9.0 PM.

Lead: fellow/ATK; producer, MATE and ELTE support. **T2.1** (M1–M4): agree access/backup, replication, irrigation contrasts and valid Sentinel-pixel geometry; test sensors before observations. **T2.2** (M4–M12) and **T2.3** (M13–M20): run seasons 1 and 2 using the unchanged model, actual Sentinel-2 inputs and independent physiology; record irrigation, yield and quality. **T2.4** (M20–M21): report primary performance before separately evaluated recalibration.

D2.1 dataset/prediction log (M20); D2.2 transfer/uncertainty report (M21). K5: two seasons; scheduled versus valid observation opportunities and biological sample size reported. K6: season-level and pooled discrimination, calibration, false alarms, lead time and transfer gap, with uncertainty or non-estimability explained. Gates: MS1–MS3.

WP3 — DrR web MVP and readiness | M1–M24 | 6.0 PM.

Lead: fellow/ATK; Krumatic engineering; AgroVIR requirements. **T3.1** (M1–M3): agree background/IP, user workflows, interfaces and acceptance tests. **T3.2** (M4–M20): build ingestion, provenance and uncertainty displays using test data/research-only functions. **T3.3** (M21–M24): integrate MS3-supported functions and test scientific consistency and invalid-input behaviour. **T3.4** (M22–M24): package the release and placement protocol.

D3.1 web MVP and D3.2 readiness dossier (M24). K7: one MVP, all agreed critical tests pass and zero unresolved critical defects in the released scope. K8: source access, rights, evidence limits, guidance and evaluation scope documented. Gate: MS4.

WP4 — AgroVIR placement | M25–M30 | 5.4 PM.

Lead: fellow with AgroVIR operational supervision; ATK oversight. **T4.1** (M25–M27): assess data requirements, output consistency, missing/out-of-domain behaviour, usability, interoperability and decision relevance. **T4.2** (M28–M30): retest priority issues and record limitations. **T4.3** (M27–M30): define the post-MSCA integration/exploitation route using approved test/coded data.

D4.1 external assessment and D4.2 roadmap (M30). K9: all six dimensions assessed and priority issues dispositioned. K10: roadmap states owners, IP/access, technical dependencies, actors, resources and decision gates. Gates: MS5/MS6.

WP5 — Training, open science, dissemination and management | M1–M30 | 4.6 PM.

Lead: fellow/ATK; supervisors and host support services contribute. **T5.1:** CDP, quarterly reviews, technical/transferable skills, knowledge exchange. **T5.2:** data/access rules from M1; DMP. **T5.3:** dissemination/communication/exploitation from M1; initial strategy at M3. **T5.4:** monthly progress/risk monitoring, agreements and reporting (all M1–M30).

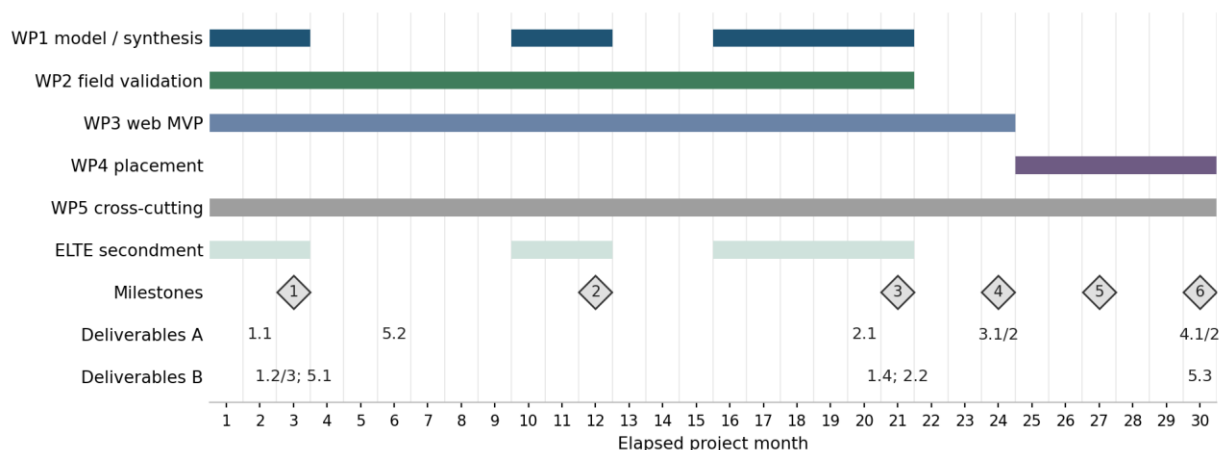
D5.1 CDP (M3); D5.2 DMP (M6; review M21/M30); D5.3 final dissemination/exploitation plan (M30; update M24). K11: CDP reviews, four technical competence strands and 12 secondment months evidenced. K12: all released assets have metadata and access decisions. K13: ≥ 2 manuscripts submitted, ≥ 2 conference presentations and ≥ 1 end-user workshop by M30. K14: each result family has an owner, audience, route and indicators.

Decision gates and dependencies. MS1 (M3): protocol lock precedes fitting and model freeze; field launch plan/CDP agreed. MS2 (M12): season 1 QC and logistics review, retaining the primary model. MS3 (M21): two-season transfer evidence and specification accepted; permitted MVP functions decided. MS4 (M24): critical tests and placement scope accepted. MS5 (M27): midpoint assessment/corrections agreed. MS6 (M30): external assessment and roadmaps completed. The fellow and ATK approve scientific gates with specialist input; AgroVIR co-reviews MS4–MS6.

Critical path: MS1 model freeze → WP2 prospective evidence → MS3 permitted functions → MS4 release → WP4 evaluation. The plan assumes a February start: MS1 in April (M3) precedes the first principal field observations in May (M4), and the second harvest falls by M20 (September). If the start slips, a feasible start is chosen within the permitted grant window before launch; field readiness is confirmed before interventions; a backup site is used only if genuinely available; if two complete seasons cannot fit, the action is formally rephased or the evidence claim narrowed, never another dataset relabelled as the primary test. WP effort totals 30.0 fellow PM: 24.0 in M1–M24 and 6.0 in M25–M30 (WP4 5.4; WP5 0.6). Pre-freeze WP1 effort is 2.2 PM of the 3.0 PM in M1–M3 (0.8 PM set up WP2/WP3/WP5); the remaining 2.8 PM are later transfer diagnostics and specification work (M10–M12, M16–M21), and standalone training, communication and software work stays out of the M16–M21 field season. Training within scientific tasks is counted once. Partner support is additional to these fellow PM.

Risk control. Monthly monitoring and gate reviews record triggers, owners and corrective decisions. Poor transfer is a reportable scientific result; it does not authorise unsupported advice.

Risk; likelihood/impact	Trigger response and responsible actors
R1 WP1 M/H	Archive rights/coverage insufficient at the M1 audit, or pipeline not reproducible by M3: use only the legally accessible common subset; freeze the specified baseline; never fit to prospective outcomes; redesign O1–O2 if the core archive is absent. Owner: fellow/ATK/MATE.
R2 WP2 M/H	Site geometry, irrigation isolation or two-season access unavailable at the field-readiness decision: activate an identified comparable site only if genuinely available; otherwise rephase or narrow the evidence claim before planting. Owner: fellow/ATK/producer.
R3 WP2 H/M	Failed calibration, insufficient interior coverage or too few matched dates: spares and manual reference checks; logged missingness; claims limited to valid coverage; UAV supplementary only. Owner: fellow/ELTE/producer.
R4 WP2–3 M/H	Poor transfer: report unchanged-model performance first; evaluate recalibration separately; limit MVP and placement to supported functions; a negative result still fulfils the reporting objective. Owner: fellow/ATK/AgroVIR.
R5 WP1–2 L/H	Prolonged supervisor absence, or a critical review or ELTE access window missed: a designated deputy and remote joint review; substitute facility only where genuinely equivalent and available. Owner: ATK/ELTE.
R6 WP3–5 M/M	Critical tests missed, source rights unresolved or placement withdrawn: scope and rights agreed before paid development; narrow to the lean MVP; replacement provider only if secured; placement change resolved with the host. Owner: fellow/ATK; Kromatic/AgroVIR.



3.2 *Quality and capacity of the host institutions and participating organisations, including hosting arrangements*

The sole beneficiary and recruiting organisation is HUN-REN ATK (facilities: Part B-2 Section 5). Required data-use, service, secondment-access and field-operation arrangements are completed before the corresponding activity or access begins, and none is claimed as signed or held. Resources are prioritised in this order: independent physiological reference and replicated field design; soil-water, weather and irrigation monitoring; essential spectral/EO acquisition; the minimum scientifically valid web MVP; optional UAV work, interface polish, extra travel or events last.

Organisation	Capacity and arrangements
HUN-REN ATK – beneficiary, recruiting organisation and academic host	HUN-REN research organisation, Martonvásár, with the fellowship anchored in the Agricultural Institute. Prof. Tibor Janda is the primary supervisor (plant-stress physiology, phenotyping, abiotic stress); Dr Roland Hollós supports the predictive-modelling/ML strand. Hosting plan: the fellow joins the Plant Physiology and Metabolomics Department as an employed researcher; M1 induction covers workspace, laboratory and instrument access, a workstation with secure versioned project storage and remote access to EO processing during ATK months; HR onboarding, reimbursement, data/IP and grants support come from the host's administration, each in place before first use.
ELTE – 12-month research secondment host	Institute of Cartography and Geoinformatics: hyperspectral sensing, geoinformatics, Earth observation and sensing-scale transfer. Prof. András Jung is the co-supervisor and secondment supervisor. The 12-month research secondment is distributed across the 24-month fellowship : band-integration and QC notebook (M1–M3), transfer diagnostics (M10–M12) and the paired-sensing clinic (M16–M21), with instrument and operator access arranged before each block. Dr Zsófia Varga supports targeted UAV acquisition where required and feasible; UAV observations remain supplementary and do not replace Sentinel-2 in the primary transferability assessment.
MATE – historical-data and irrigation-science contributor	Provides the 2022–2026 controlled-irrigation processing-tomato archive , including soil-water, physiological, spectral, meteorological and agronomic observations. Dr Sándor Takács contributes processing-tomato irrigation and AquaCrop expertise, while Prof. Gábor Milics contributes precision-agriculture expertise. An M1 archive and rights audit fixes the accessible years, variables and permitted analysis; only the legally accessible common subset is used, and redistribution or commercial reuse is not assumed.
Krumatic EOOD – remunerated software-engineering support	Krumatic provides paid specialist software-engineering services for implementation of the DrR web MVP from the fellow's pre-existing desktop research prototype; scientific specifications, model validation and interpretation remain with the fellow/HUN-REN ATK. A deliverable-based scope (ingestion, provenance, frozen-model inference, uncertainty display, acceptance tests, source delivery and rights) is contracted before paid development; a lean MVP contingency defers dashboards and production FMIS integration.
AgroVIR Kft. – non-academic placement host	Hungarian FMIS company. AgroVIR provides operational requirements and feedback during the fellowship and hosts the M25–M30 non-academic placement , where the validated DrR web MVP is evaluated for usability, decision relevance, data requirements and interoperability. Proprietary customer-data access is not a project dependency.
Commercial farmer/producer – prospective field-trial host and validation environment	An independent Hungarian farmer/producer, irrespective of gender , provides the commercial field for two prospective seasons and implements the agreed treatments and routine management under the field protocol. Adopted requirements, decided at the field-readiness gate before planting: 12 separately managed treatment areas with independently controllable water delivery, Sentinel-2-valid interior cores and two-season availability; no site is described as secured. The project installs soil-water and meteorological monitoring and conducts all specialised observations; design, validation, analysis and interpretation remain with the fellow/HUN-REN ATK.
MVCRI, Plovdiv – post-MSCA continuation route	Prof. Ganeva supports a potential post-MSCA continuation route at MVCRI for crop-specific recalibration and transfer to Bulgarian production conditions; it is not a guaranteed position, funded activity or multi-crop validation programme. MVCRI has no core in-action validation role during the project.